Lab 05 Derrick K. Gibbs-McGlaston

This lab helped me understand why preparing data is such an important part of machine learning. I have heard many times that the quality of the data affects the quality of the model, but this lab gave me a chance to see what that means in practice. The data cannot always be used exactly as it is given. There are steps that have to be taken before it is ready for a machine learning model.

The first part that stood out to me was dealing with missing values in the Age column. Instead of removing every passenger who had a missing age, I used the median age to fill in those missing values. This helped me understand that missing data does not always mean that the entire row has to be thrown away. There are ways to handle missing information while keeping the rest of the useful data.

Using the median also made sense after thinking about what it represents. The median gives a middle value and is not affected as much by extremely high or low values. After filling the missing ages, I could check the data again and see that those missing values had been handled. That made this part of the process feel less theoretical because I could actually see the change in the dataset.

The next part of the lab involved one-hot encoding the Sex and Embarked columns. This was probably one of the most useful parts for me because it showed me how information that makes sense to a person has to be changed into a form that a machine learning model can use. A person can easily understand words such as male, female, or the name of a location. A model needs those categories represented numerically.

Before doing this lab, I could have looked at encoding as simply changing words into numbers. I understand it better now. The goal is to represent categories in a numerical form without creating a relationship between them that does not really exist. Seeing the new columns after encoding helped me understand what was actually happening to the data.

Standardizing Age and Fare added another part to the preparation process. Those two features use very different ranges of numbers. Fare can contain much larger values than age. Standardization puts numerical features onto a more comparable scale instead of allowing the size of the original numbers to have too much influence.

This lab also helped me see how the different parts of the machine learning workflow connect. In the previous lab, I explored the Titanic data and looked for patterns and relationships. In this lab, I worked with the same type of data from a different point of view. Instead of asking what the data was showing me, I was preparing the data so that it could eventually be used by a model.

That connection made the overall process clearer to me. Exploratory Data Analysis helps me understand what is in the dataset and identify possible problems. Data preparation is where I begin doing something about those problems. Missing values have to be handled, categorical information has to be converted, and numerical features may need to be placed on a similar scale.

Another thing I learned is that data preparation is not just a list of steps that should automatically be applied to every dataset. There is a reason behind each decision. I need to understand the data well enough to know why I am filling a missing value, why I am encoding a particular feature, or why certain numerical values need to be standardized.

After completing this lab, the machine learning workflow makes more sense to me. I can see that building the actual model is only one part of the process. A lot of important work happens before the model is ever trained. Preparing the data correctly gives the model better information to work with and makes the later stages of the workflow more reliable.